ENG3014 Communications Systems 3: Communication Channel Simulation Report

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1 Introduction

This report presents the digital communication using binary frequency shift keying [FSK]. The user define ASCII message to be converted into a binary modulated into a carrier, transmitted through a simulated channel with noise and fading option. The system preformence is pased on using the bit error rate [BER] across multiple retransmissions and thre SNR conditions (6, 18, and 24 dB). The similation also include bandwidth limiting to apply Hamming [7,4] error correction. The gole is to demonstrate a complete chain transmiting channel and analyze how the channel weekness and design choices affect its reliability.

2 Background

Modulation Formats

Digital modulation encodes bits into waveforms that can be transmitted over a channel. In binary amplitude shift keying [ASK], the bit value changes the carrier amplitude; in binary phase shift keying [PSK], it changes the carrier phase; and in binary frequency shift keying [FSK], it changes the carrier frequency. [FSK] is especially prone to noisy channels because its detection relies on frequency "energy in two tones" rather than amplitude, making it less sensitive to amplitude fluctuations. In this project, binary FSK is used, with each bit selecting one of two tones around a carrier: $f_0 = f_c - \Delta f$ **and** $f_1 = f_c + \Delta f$. This keeps the system simple while demonstrating a complete modulation/demodulation of channel.

Fading Model

To imitate an urban wireless channel, the simulation applies slow amplitude fading. A random Gaussian noise is smoothed with a moving average window to generate a low frequency envelope that varies around 1.0. The fading levels scales the depth of the fades, while the fading rate controles how quickly the envelope change over time. This illistrate realistic gradual dips

and rises in the signal strength, similar to shadowing and multipath effects apsorved in city enviroments.

Addition: Hamming (7,4)

Hamming (7,4) coding adds error correction to the messeage by expanding every 4 data bits into a 7 bit code word using three parity bits. At the receiver, a syndrome is computed to identify and correct any single bit error within each code word. This improves reliability, especially at low SNR, at the cost of additional bandwidth and reduced effective data rate. In the simulation, Hamming coding can be enabled to compare BER with and without error correction.

3 Simulation Approach

Message and Bitstream: A user defined ASCII message is converted to a binary array and repeated until it exceeds 2048 bits.

Modulation: Bits are mapped to two tones ($f_c \pm \Delta f$). The sampling frequency is set by the symbol baud rate R_b and samples per symbol *sps*: $f_s = R_b \cdot \text{sps}$. So if R_b symbols per second been send and take *sps* samples per symbol, the total samples per second is their product.

Channel: The signal is optionally faded, corrupted with AWGN at 6/18/24 dB SNR, and bandlimited to $B/f_c \leq 0.2$ using an FFT domain mask.

Demodulation: The receiver correlates the signal with local oscillators at f_0 and f_1, integrates energy per symbol, and selects the larger energy as the detected bit.

BER Measurement: The same message is retransmitted multiple times with independent noise. BER is averaged across runs to stabilize results.

4 Results

Table 1 summarizes BER values across SNR levels. At higher SNR, the BER falls not as expected. The Hamming (7,4) option should provides additional improvement, the table values shows the BER stays fairly high and does not change much as SNR increases. Hamming (7,4) only gives a small improvement. Which means the main problem is not noise, but could be the receiver struggling to separate the two tones. Using a larger (Δf) more samples per symbol or a better detector would likely reduce BER.

SNR (dB)	BER (no coding)	BER (Hamming 7,4)
6	0.4311	0.4291
18	0.4178	0.4178
24	0.4178	0.4178

Table 1: Tuned configuration (fc=1000, df/Rb=1.0, sps=80), fading OFF.

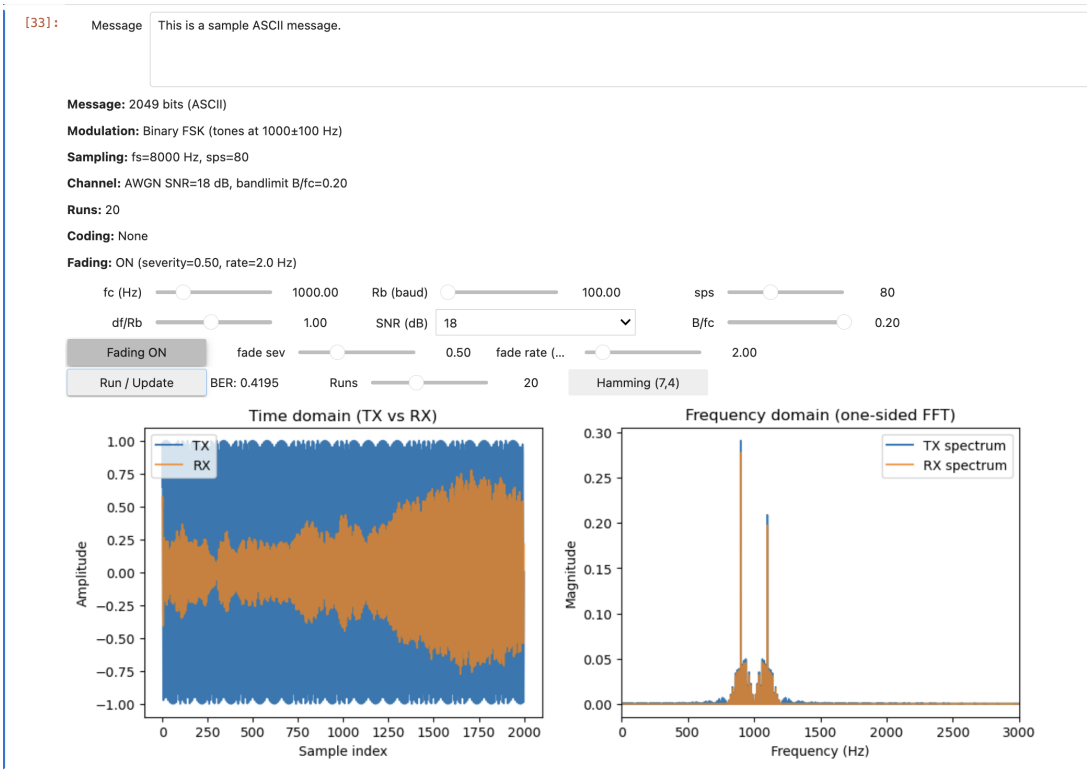


Figure 1: Time domain TX/RX waveforms and FSK(fc=1000 Hz, Δf=100 Hz) showing channel distortion and tone separation.

5 Discussion and Conclusions

The simulation meets the project requirements: message encoding, FSK modulation, AWGN channel at 6/18/24 dB, bandwidth limiting, demodulation, and BER analysis across retransmissions. Results show BER is sensitive to tone spacing and sampling resolution; when (Δf) is small, the two tones overlap and errors remain high even at higher SNR. Hamming (7,4) provides only a modest improvement in this setting, indicating that detection limitations dominate. Overall, the results suggest that improving tone separation or receiver design would yield the largest BER gains.

References

[1] Lavery, M. P. J., *ENG3014 Communications Systems 3 – Lecture Notes and Laboratory Materials*, University of Glasgow, 2025/26.

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